

# Storage

Storage holds data, instructions, and information for future use

A **storage medium** is the physical material on which a computer keeps data, instructions, and information

# Storage



# Storage

- **Capacity** is the number of bytes a storage medium can hold

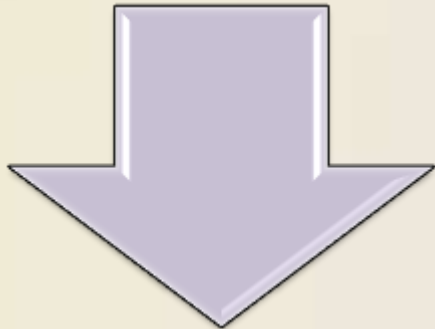
| Storage Terms  |                             |                                               |
|----------------|-----------------------------|-----------------------------------------------|
| Storage Term   | Approximate Number of Bytes | Exact Number of Bytes                         |
| Kilobyte (KB)  | 1 thousand                  | $2^{10}$ or 1,024                             |
| Megabyte (MB)  | 1 million                   | $2^{20}$ or 1,048,576                         |
| Gigabyte (GB)  | 1 billion                   | $2^{30}$ or 1,073,741,824                     |
| Terabyte (TB)  | 1 trillion                  | $2^{40}$ or 1,099,511,627,776                 |
| Petabyte (PB)  | 1 quadrillion               | $2^{50}$ or 1,125,899,906,842,624             |
| Exabyte (EB)   | 1 quintillion               | $2^{60}$ or 1,152,921,504,606,846,976         |
| Zettabyte (ZB) | 1 sextillion                | $2^{70}$ or 1,180,591,620,717,411,303,424     |
| Yottabyte (YB) | 1 septillion                | $2^{80}$ or 1,208,925,819,614,629,174,706,176 |

# Storage

- A **storage device** is the computer hardware that records and/or retrieves items to and from storage media



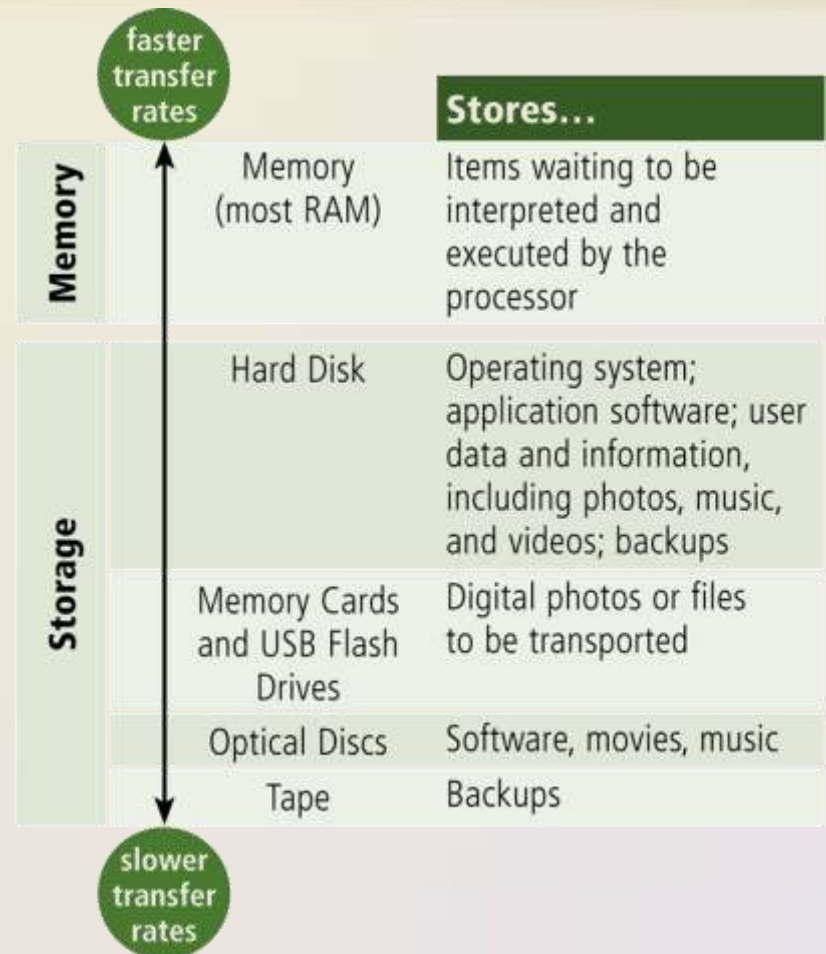
**Reading** is the process of transferring items from a storage medium into memory



**Writing** is the process of transferring items from memory to a storage medium

# Storage

- **Access time** measures:
  - The amount of time it takes a storage device to locate an item on a storage medium
  - The time required to deliver an item from memory to the processor



# Hard Disks

- A **hard disk** contains one or more inflexible, circular platters that use magnetic particles to store data, instructions, and information



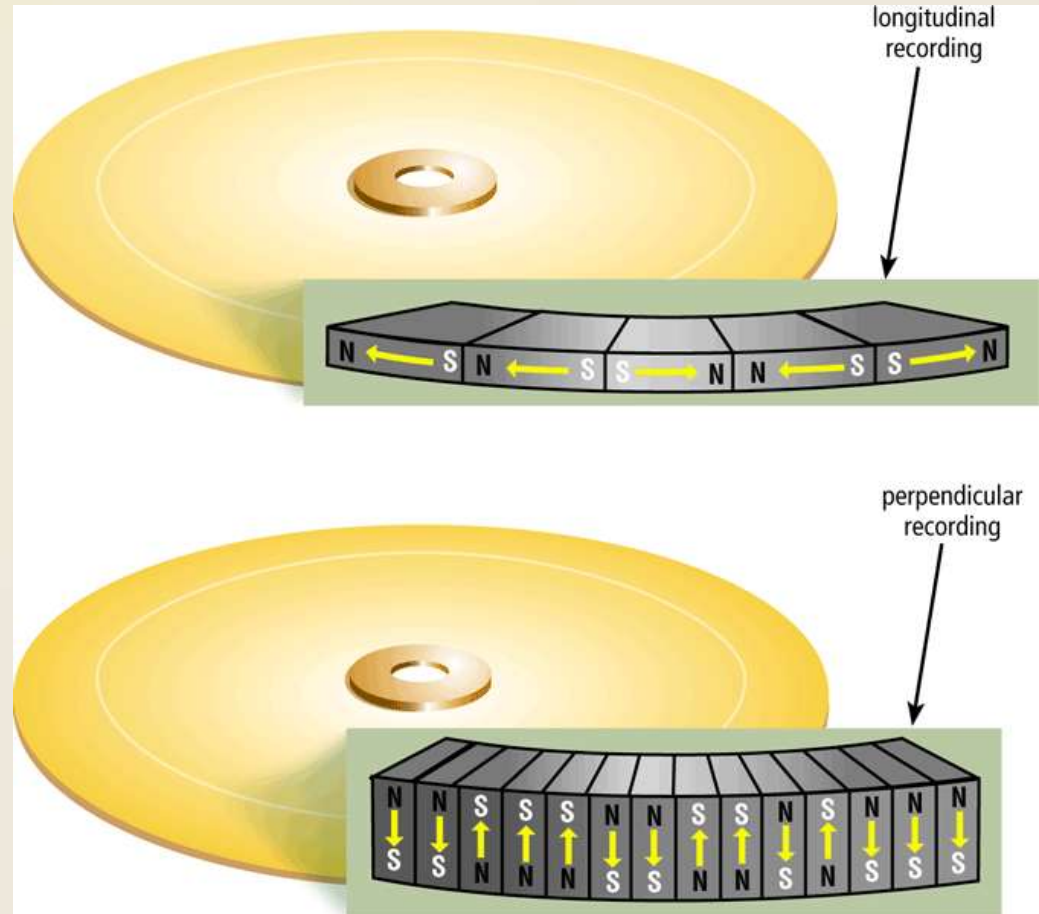
hard disk  
mounted inside  
system unit

close-up of  
hard disk



# Hard Disks

- Hard disks can store data using longitudinal recording or perpendicular recording



# Hard Disks

- Characteristics of a hard disk include:

Capacity

Platters

Read/Write  
Heads

Cylinders

Sectors and  
Tracks

Revolutions  
per Minute

Transfer  
Rate

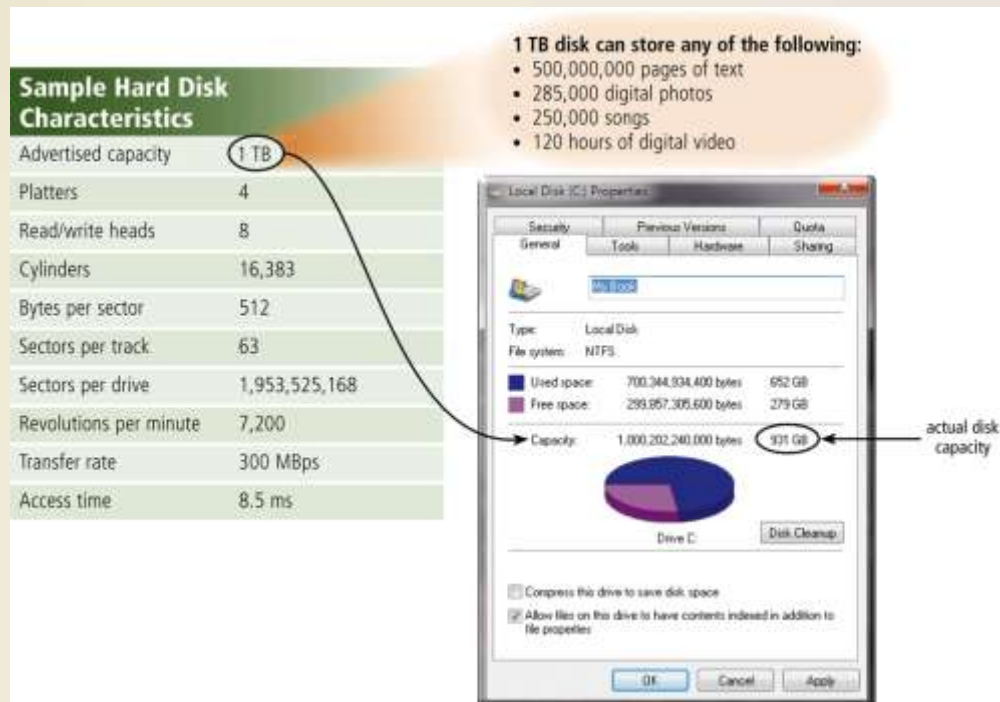
Access Time





# Hard Disks

- **Formatting** is the process of dividing the disk into tracks and sectors so that the operating system can store and locate data and information on the disk



# Hard Disks

## How a Hard Disk Works

### Step 1

The circuit board controls the movement of the head actuator and a small motor.

### Step 2

A small motor spins the platters while the computer is running.

### Step 3

When software requests a disk access, the read/write heads determine the current or new location of the data.

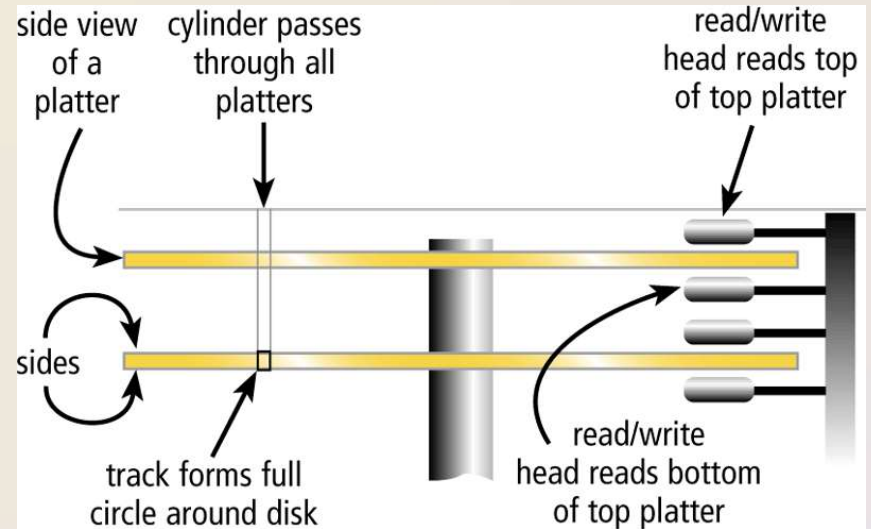
### Step 4

The head actuator positions the read/write head arms over the correct location on the platters to read or write data.



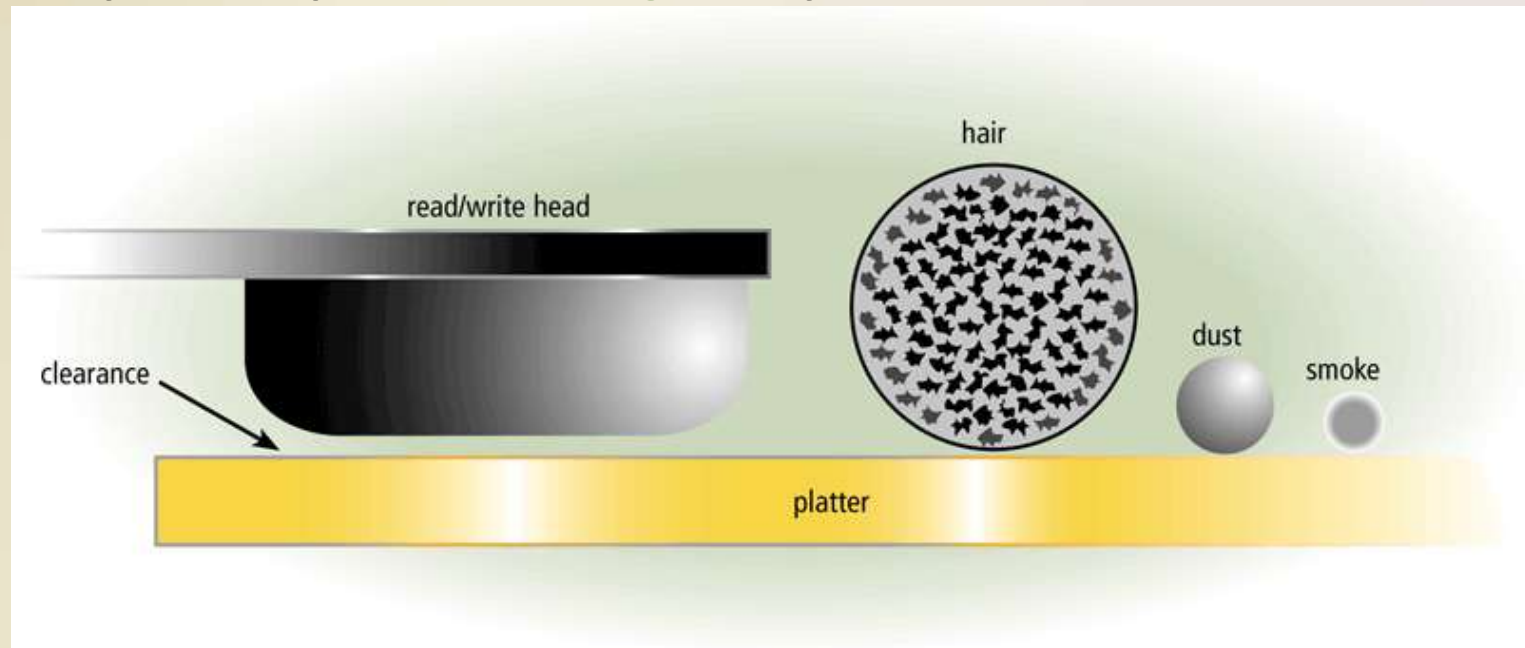
# Hard Disks

- The hard disk arms move the read/write head, which reads items and writes items in the drive
  - Location often is referred to by its cylinder



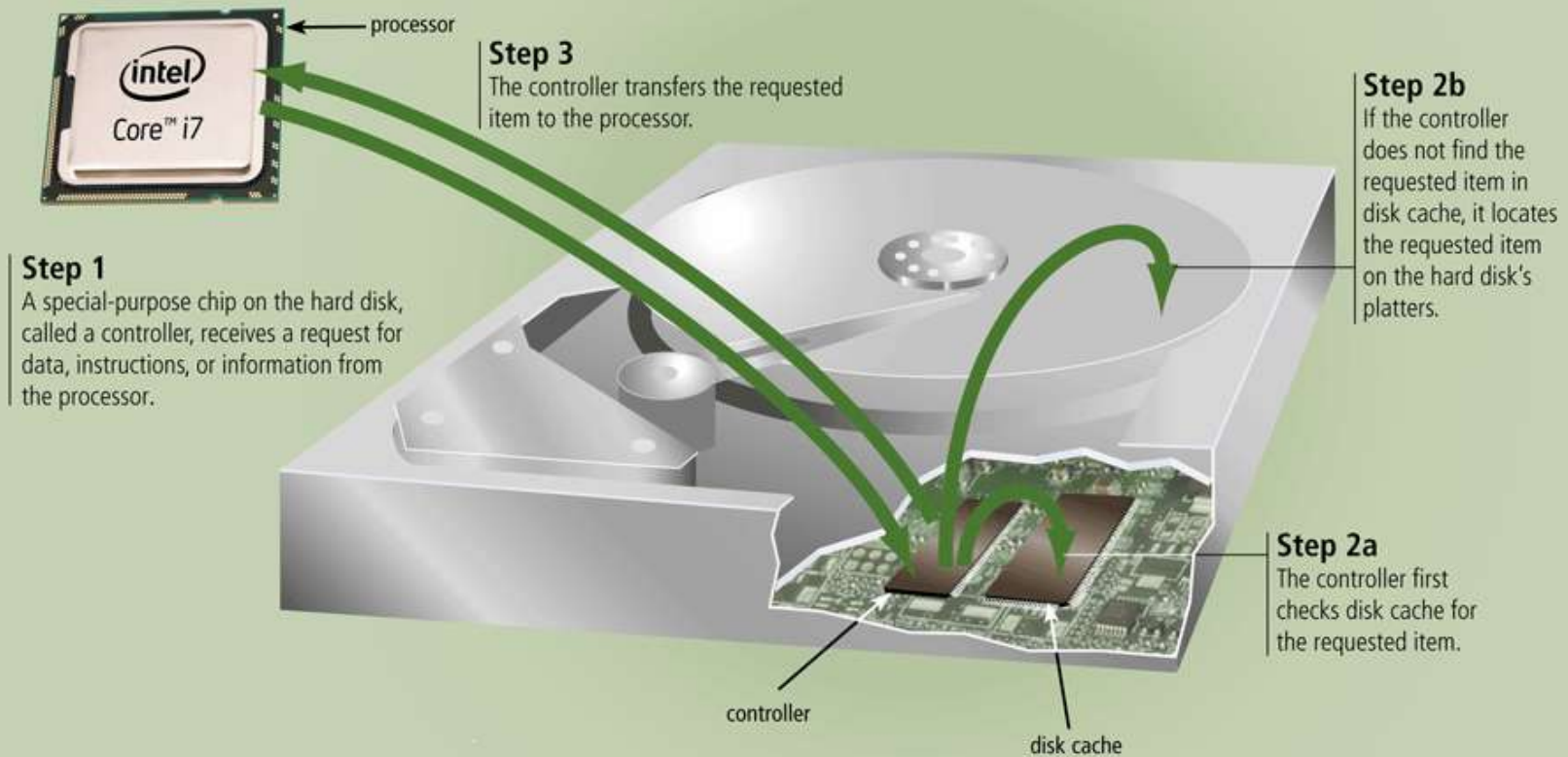
# Hard Disks

- A head crash occurs when a read/write head touches the surface of a platter
- Always keep a **backup** of your hard disk



# Hard Disks

## How Disk Cache Works



# Hard Disks

- **RAID** (redundant array of independent disks) is a group of two or more integrated hard disks
- A **network attached storage** (NAS) device is a server connected to a network with the sole purpose of providing storage





# Hard Disks



An **external hard disk** is a separate free-standing hard disk that connects to your computer with a cable or wirelessly



A **removable hard disk** is a hard disk that you insert and remove from a drive



Internal and external hard disks are available in miniature sizes (miniature hard disks)

# Hard Disks

- A disk controller consists of a special-purpose chip and electronic circuits that control the transfer of data, instructions, and information from a disk to and from the system bus and other components of the computer

**SATA**

**EIDE**

**SCSI**

**SAS**

# Flash Memory Storage

- Flash memory chips are a type of solid state media and contain no moving parts
- **Solid state drives (SSDs)** have several advantages over magnetic hard disks:

Faster access time

Faster transfer rates

Generate less heat and  
consume less power

Last longer

# Flash Memory Storage



# Flash Memory Storage

- A **memory card** is a removable flash memory device that you insert and remove from a slot in a computer, mobile device, or card reader/writer

CompactFlash  
(CF)

Secure Digital  
(SD)

Secure Digital  
High Capacity  
(SDHC)

microSD

microSDHC

xD Picture  
Card

Memory Stick

Memory Stick  
Micro (M2)

# Flash Memory Storage



## Various Memory Cards

| Media Type              |                                                                                       | Storage Capacity | Use                                                                                                                |
|-------------------------|---------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------|
| CompactFlash (CF)       |    | 512 MB to 100 GB | Digital cameras, smart phones, PDAs, photo printers, portable media players, notebook computers, desktop computers |
| Secure Digital (SD)     |    | 512 MB to 8 GB   | Digital cameras, digital video cameras, smart phones, PDAs, photo printers, portable media players                 |
| SDHC                    |    | 4 to 32 GB       | Digital cameras                                                                                                    |
| microSD                 |    | 1 to 2 GB        | Smart phones, portable media players, handheld game consoles, handheld navigation devices                          |
| microSDHC               |    | 4 to 16 GB       | Smart phones, portable media players, handheld game consoles, handheld navigation devices                          |
| xD Picture Card         |   | 256 MB to 2 GB   | Digital cameras, photo printers                                                                                    |
| Memory Stick PRO Duo    |  | 1 to 16 GB       | Digital cameras, smart phones, handheld game consoles                                                              |
| Memory Stick Micro (M2) |  | 1 to 16 GB       | Smart phones                                                                                                       |

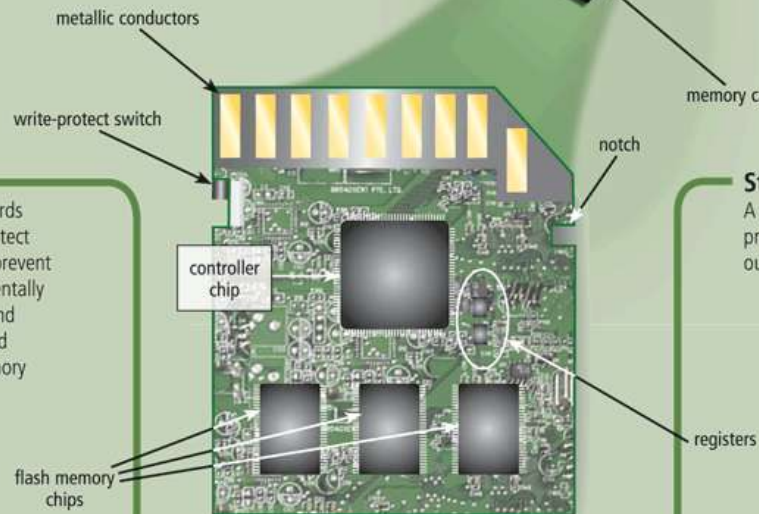


# Flash Memory Storage

## How One Type of Memory Card Works

### Step 1

When you insert a memory card in a card reader/writer or card slot, the memory card's metallic conductors make contact with connectors in the card reader/writer or card slot, allowing the transfer of photos and other items between the card and the reading/writing device.



### Step 2

A notch on the side of the memory card prevents the card from accidentally slipping out of the card reader/writer or card slot.

### Step 4

Some memory cards contain write-protect switches, which prevent users from accidentally erasing photos and other items stored on the flash memory chips.

### Step 3

Flash memory chips store photos and other types of data and information. When requested, the controller transfers items stored on the flash memory chips to the metallic conductors, using registers for temporary storage, as needed.

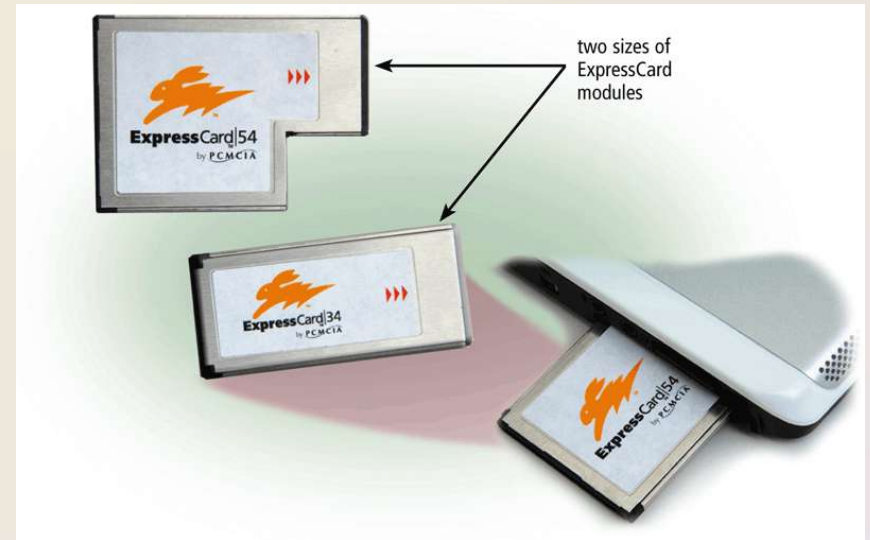
# Flash Memory Storage

- **USB flash drives** plug into a USB port on a computer or mobile device



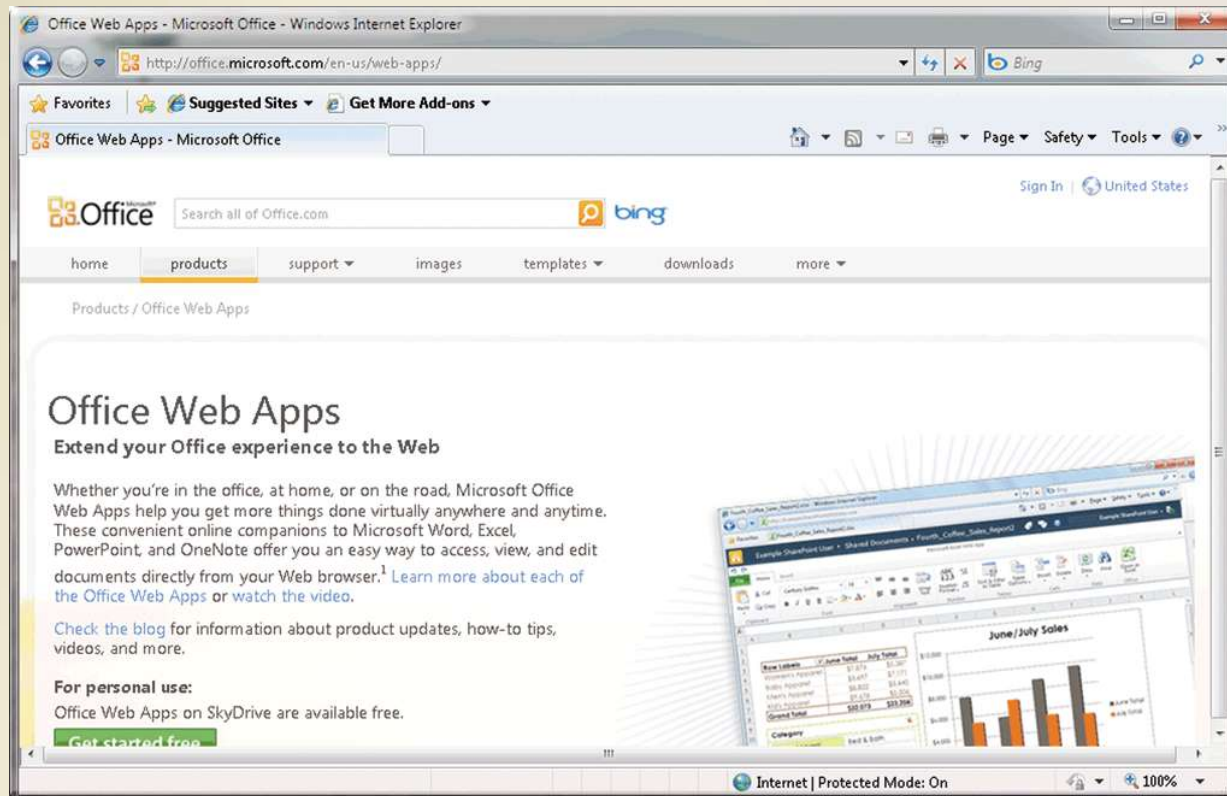
# Flash Memory Storage

- An **ExpressCard module** is a removable device that fits in an ExpressCard slot
- Developed by the PCMCIA
- Commonly used in notebook computers



# Cloud Storage

- **Cloud storage** is an Internet service that provides storage to computer users



# Cloud Storage

## Cloud Storage Providers

| Web Site Names                           | Type of Storage Provided                                           | Other Services                     |
|------------------------------------------|--------------------------------------------------------------------|------------------------------------|
| Box.net, IDrive, Windows Live SkyDrive   | Backup or additional storage for any type of file                  |                                    |
| Flickr, Picasa                           | Digital photos                                                     | Photo editing and photo management |
| YouTube                                  | Digital videos                                                     |                                    |
| Facebook, MySpace                        | Digital photos, digital videos, messages, and personal information | Social networking                  |
| Google Docs, Office Web Apps             | Documents, spreadsheets, presentations                             | Productivity suite                 |
| Gmail, Windows Live Hotmail, Yahoo! Mail | E-mail messages                                                    |                                    |
| Amazon EC2, Amazon S3, Nirvanix          | Enterprise-level storage                                           | Web services, data center services |

# Cloud Storage

- Users subscribe to cloud storage for a variety of reasons:

Access files from any computer

Store large files instantaneously

Allow others to access their files

View time-critical data and images immediately

Store offsite backups

Provide data center functions

# Optical Discs

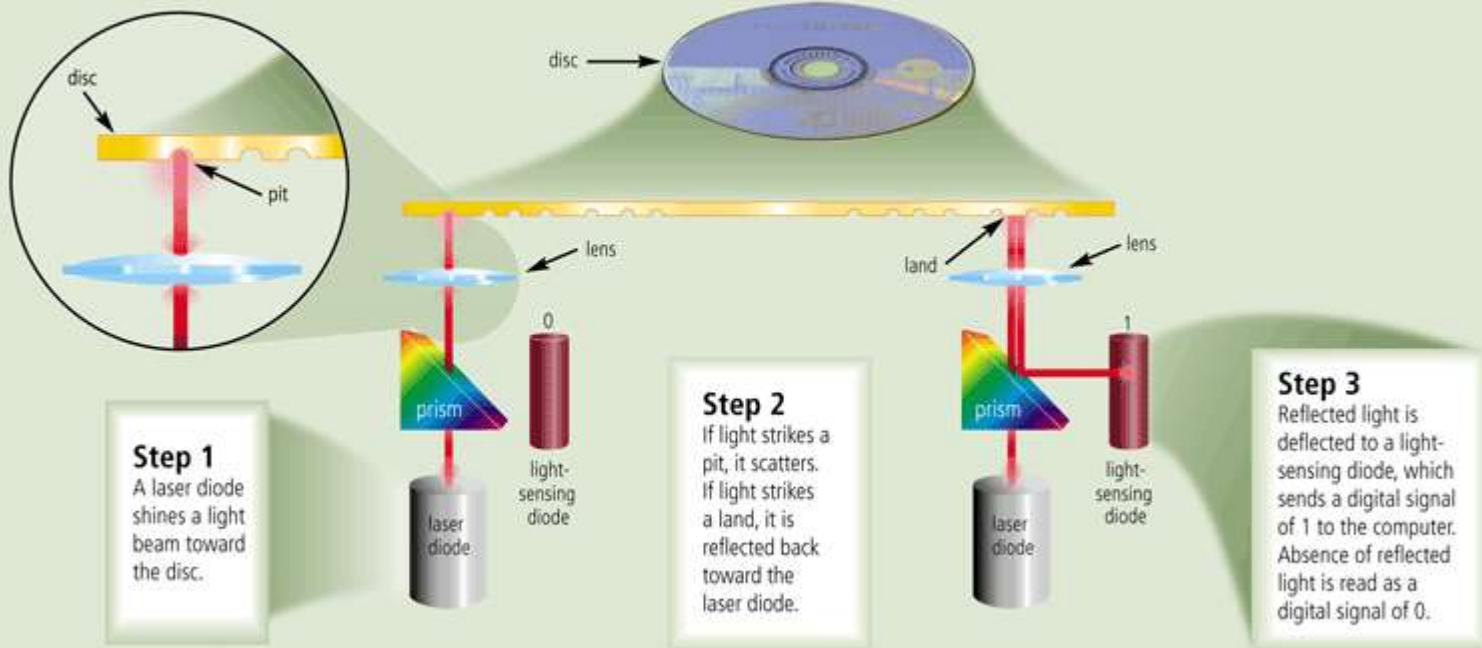
- An optical disc consists of a flat, round, portable disc made of metal, plastic, and lacquer that is written and read by a laser
- Typically store software, data, digital photos, movies, and music
- Read only vs. rewritable





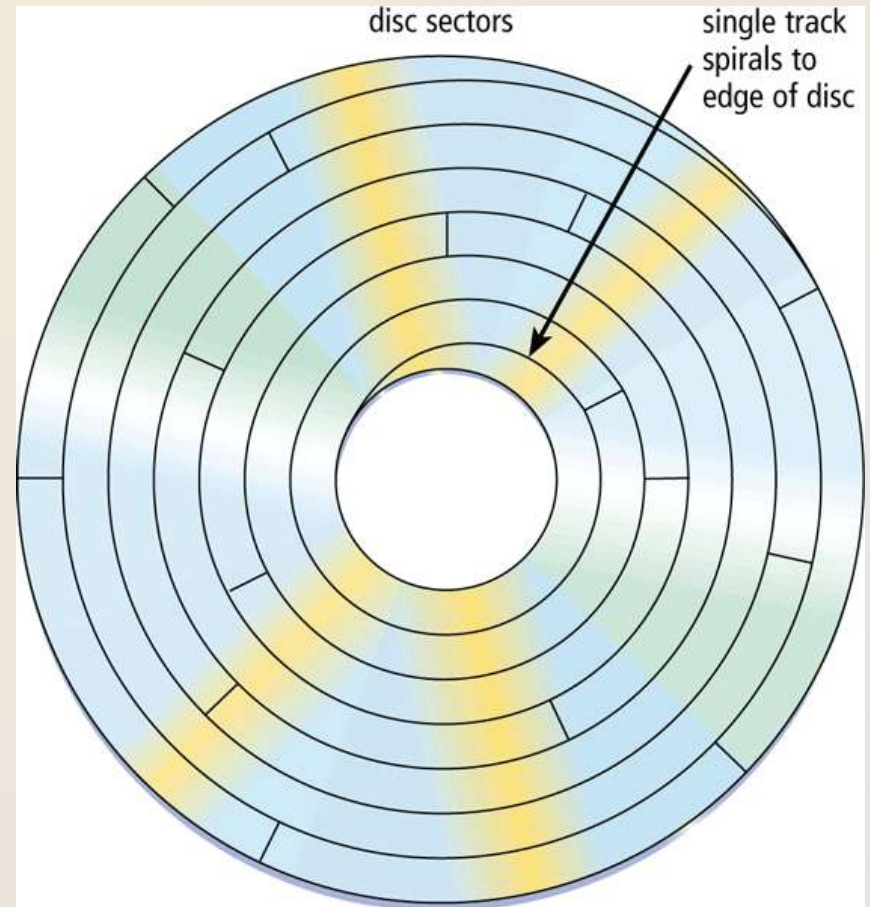
# Optical Discs

## How a Laser Reads Data on an Optical Disc



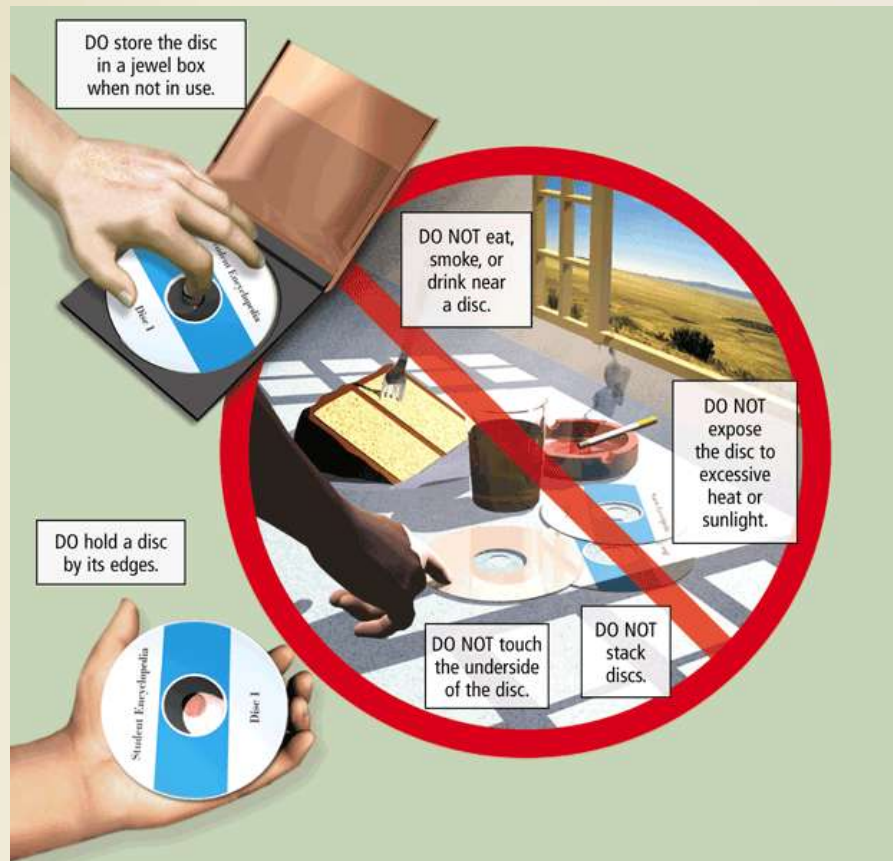
# Optical Discs

- Optical discs commonly store items in a single track that spirals from the center of the disc to the edge
- Track is divided into evenly sized sectors



# Optical Discs

- Care of optical discs



# Optical Discs



A **CD-ROM** can be read from but not written to

- Read from a **CD-ROM drive** or CD-ROM player



A **CD-R** is a multisession optical disc on which users can write, but not erase



A **CD-RW** is an erasable multisession disc

- Must have a **CD-RW drive**

# Optical Discs

## Archive disc

- Stores photos from an online photo center
- Resolution usually is 7200 pixels per photo
- Cost is determined by the number of photos being stored

## Picture CD

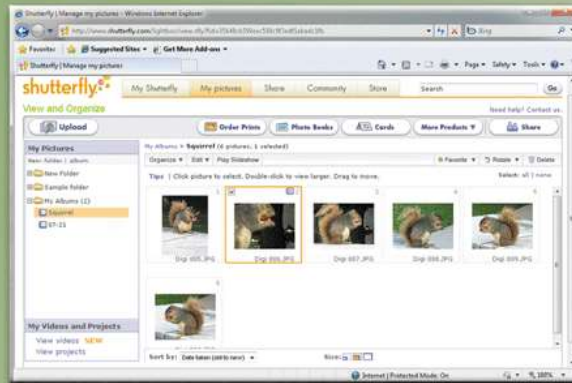
- Single-session CD-ROM that stores digital versions of film
- Typically uses a 1024 x 1536 resolution
- Many photo centers offer Picture CD services

# Optical Discs

## How an Archive Disc Works

### Step 1

Upload your digital photos to a photo sharing community for others to view.



### Step 2

Select the photos to be stored on the archive disc and then place your order.



### Step 3

Pick up your archive disc at a designated store or receive it in the mail. At home, edit and/or print images from the archive disc on your ink-jet or photo printer, or view the images on a monitor or television screen. At a store, edit and/or print images from the archive disc at a kiosk.





# Optical Discs



A **DVD-ROM** is a high-capacity optical disc on which users can read but not write or erase

- Requires a **DVD-ROM** drive



A Blu-ray Disc-ROM (BD-ROM) has a storage capacity of 100 GB



**DVD-RW, DVD+RW, and DVD+RAM** are high-capacity rewritable DVD formats



# Other Types of Storage



Tape



Magnetic stripe  
cards and smart  
cards



Microfilm and  
microfiche



Enterprise storage

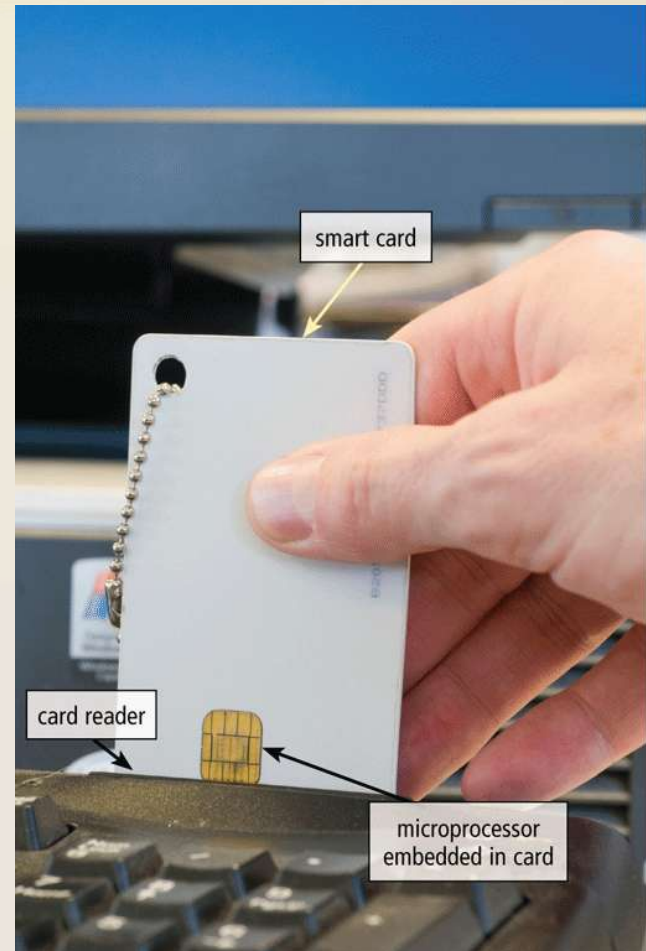
# Other Types of Storage

- **Tape** is a magnetically coated ribbon of plastic capable of storing large amounts of data and information
- A **tape drive** reads and writes data and information on a tape



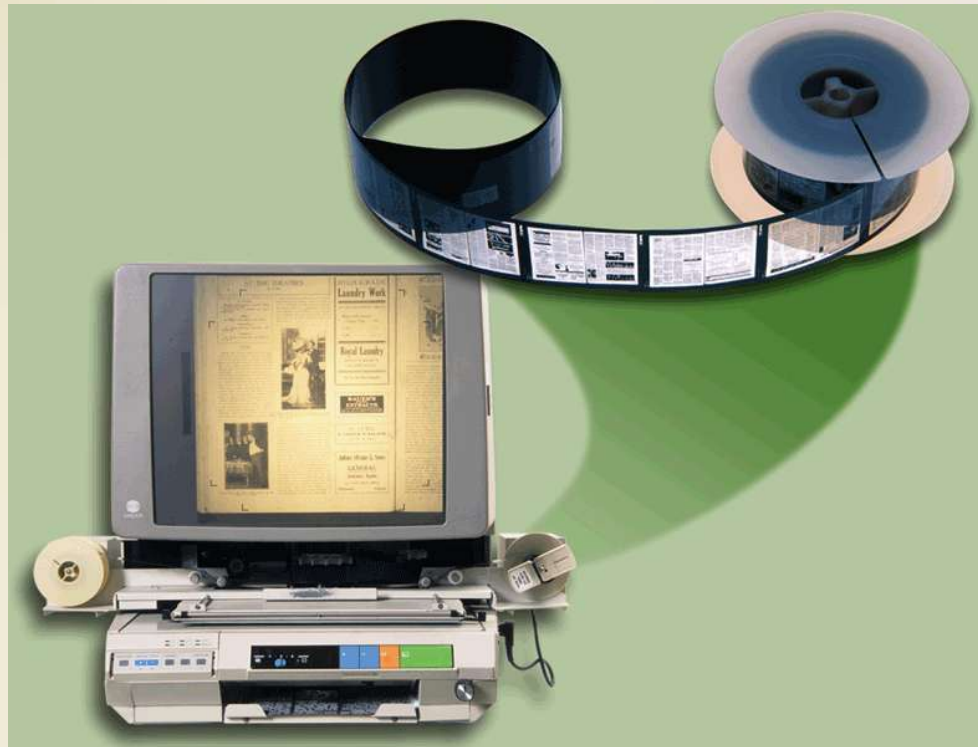
# Other Types of Storage

- A **magnetic stripe card** contains a magnetic stripe that stores information
- A **smart card** stores data on a thin microprocessor embedded in the card



# Other Types of Storage

- **Microfilm** and **microfiche** store microscopic images of documents on a roll or sheet film



# Other Types of Storage

| Media Life Expectancies* (when using high-quality media) |                            |                           |
|----------------------------------------------------------|----------------------------|---------------------------|
| Media Type                                               | Guaranteed Life Expectancy | Potential Life Expectancy |
| Magnetic disks                                           | 3 to 5 years               | 20 to 30 years            |
| Optical discs                                            | 5 to 10 years              | 50 to 100 years           |
| Solid state drives                                       | 50 years                   | 140 years                 |
| Microfilm                                                | 100 years                  | 500 years                 |
| * according to manufacturers of the media                |                            |                           |

# Other Types of Storage

- Enterprise storage stores huge volumes of data and information for large businesses
  - Uses special hardware for heavy use, maximum availability, and maximum efficiency





# Putting It All Together



## Home user

- 500 GB hard disk
- Cloud storage
- Optical disc drive
- Card reader/writer
- USB flash drive



## Small Office/Home Office user

- 1 TB hard disk
- Cloud storage
- Optical disc drive
- External hard disk for backup
- USB flash drive



## Mobile

- 250 GB hard disk
- Cloud storage
- Optical disc drive
- Card reader/writer
- Portable hard disk for backup
- USB flash drive

# Putting It All Together



## **Power User**

- 2.5 TB hard disk
- Cloud storage
- Optical disc drive
- Portable hard disk for backup
- USB flash drive



## **Enterprise User (desktop computer)**

- 1 TB hard disk
- Optical disc drive
- Smart card reader
- Tape drive
- USB flash drive



## **Enterprise User (server or mainframe)**

- Network storage server
- 40 TB hard disk system
- Optical disc server
- Microfilm or microfiche